

DOUBLE-GATE MOSFET AMPLIFIER

with 14nm technology

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INTRODUCTION

- MOSFETs are the fundamental building blocks of modern electronic devices such as processors and integrated circuits.
- They provide high switching speed and low power consumption, making them essential for modern electronics.
- As device dimensions scale into the nanometer range, conventional single-gate MOSFETs suffer from:
 - Short-channel effects
 - Increased leakage current
 - Poor gate control
- Double-Gate MOSFETs use two gates to control the channel, improving electrostatic control and reducing leakage.
- In this project, a 14 nm Double-Gate MOSFET is designed and simulated using Synopsys Sentaurus TCAD.

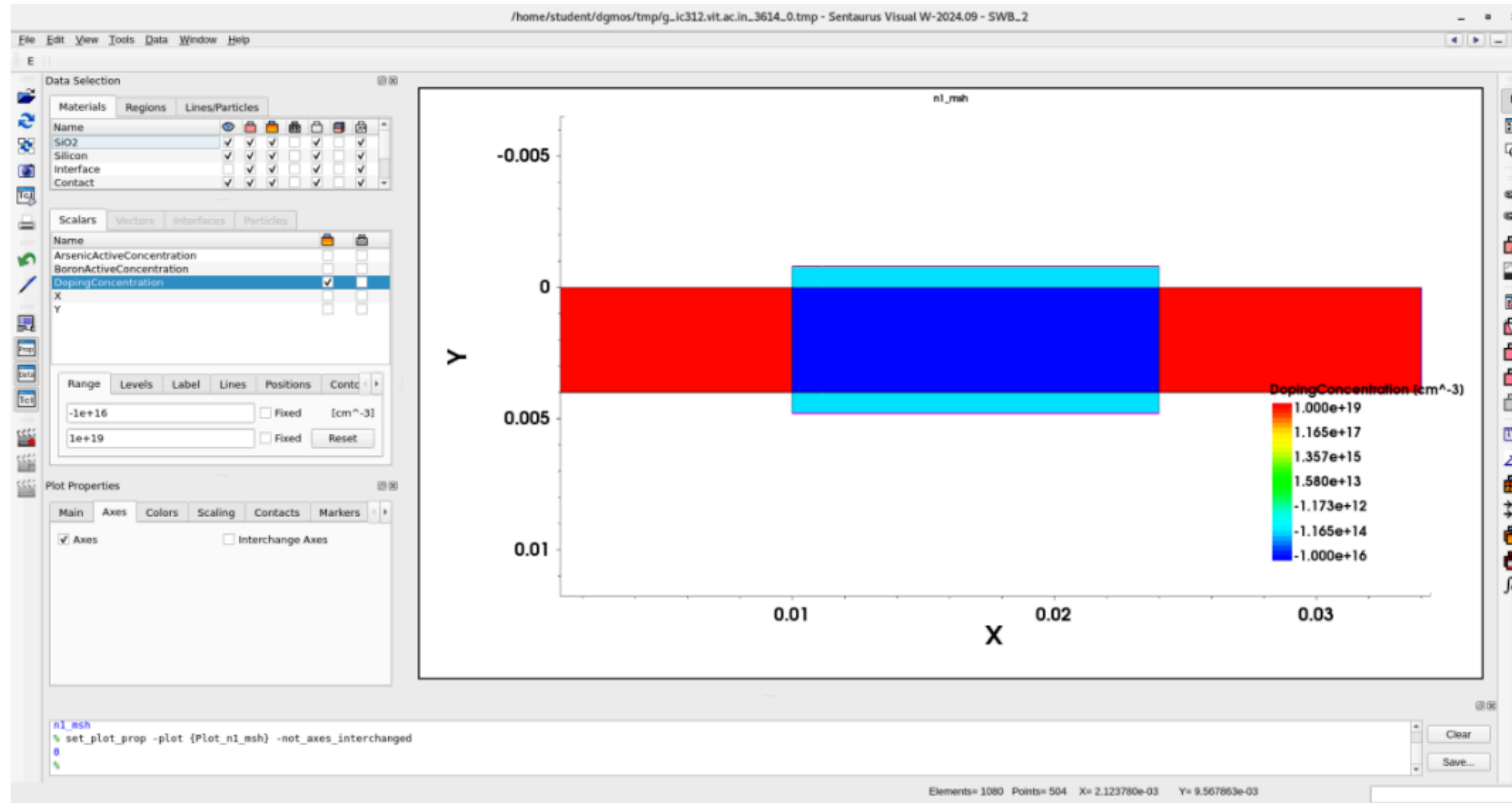
Device dimensions used in the simulation:

- Source: 10 nm
- Channel: 14 nm
- Drain: 10 nm
- Silicon thickness: 4 nm
- Oxide thickness: 1 nm

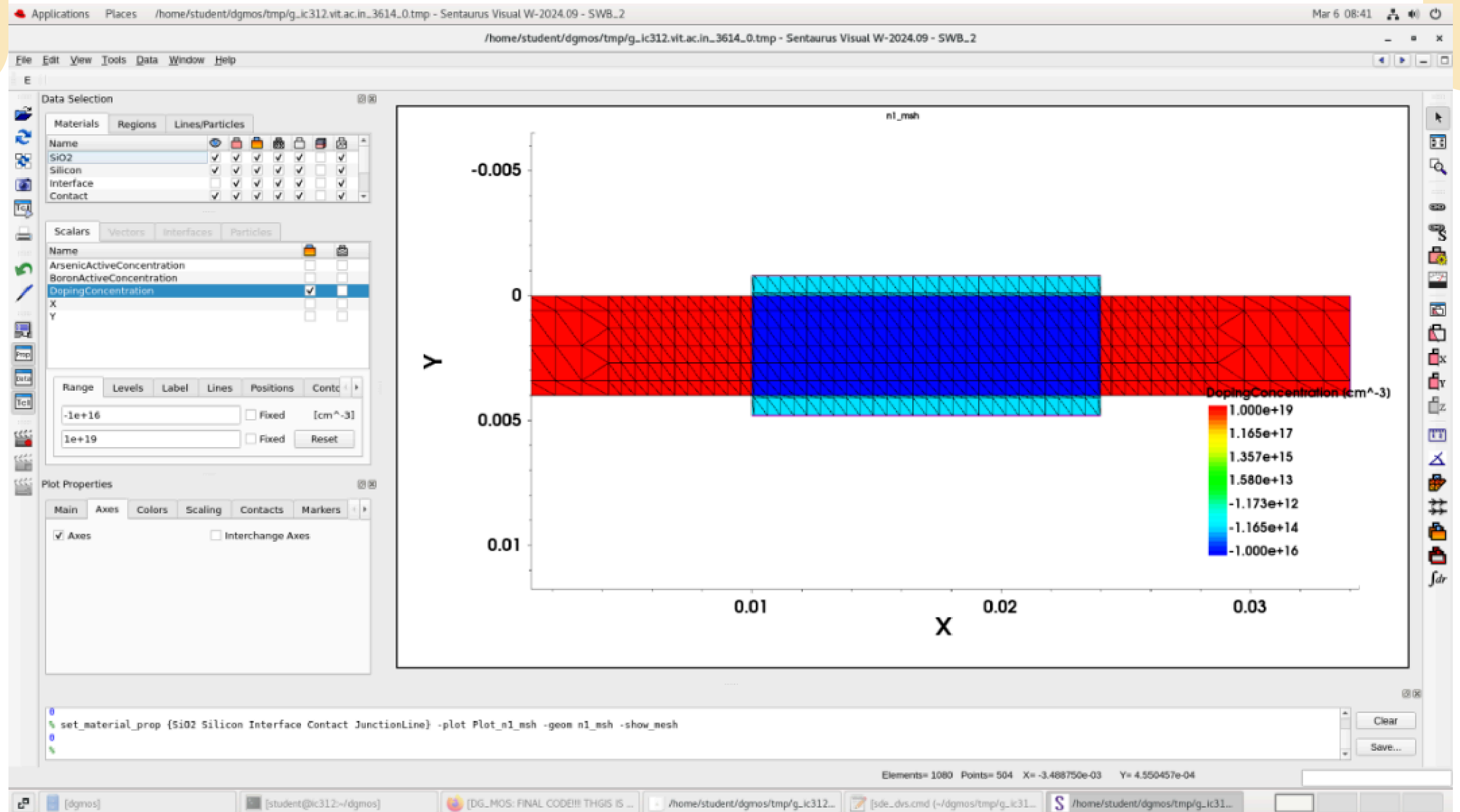
The design helps analyze short-channel behavior and improved gate control at nanoscale technology.

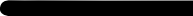
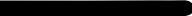
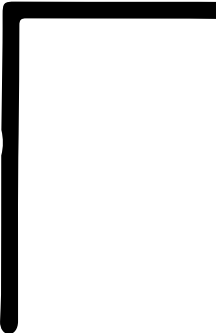
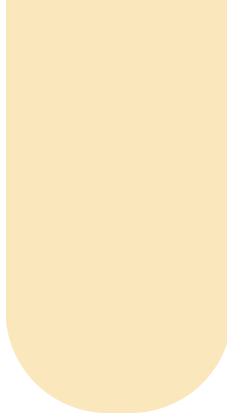
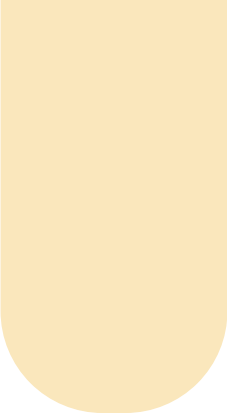
DEVICE DESIGN IN SDE

(DOUBLE GATE NMOS)



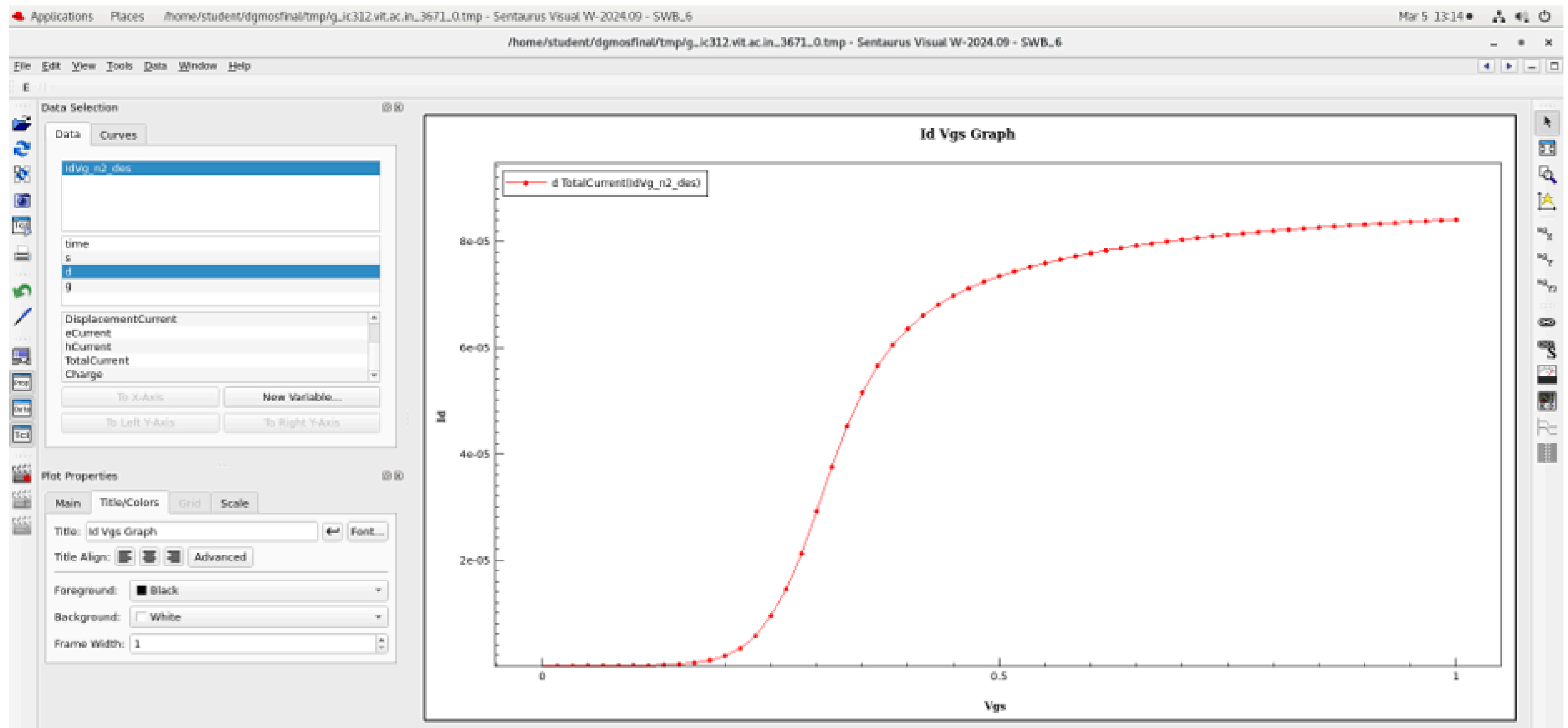
DEVICE DESIGN WITH MESH



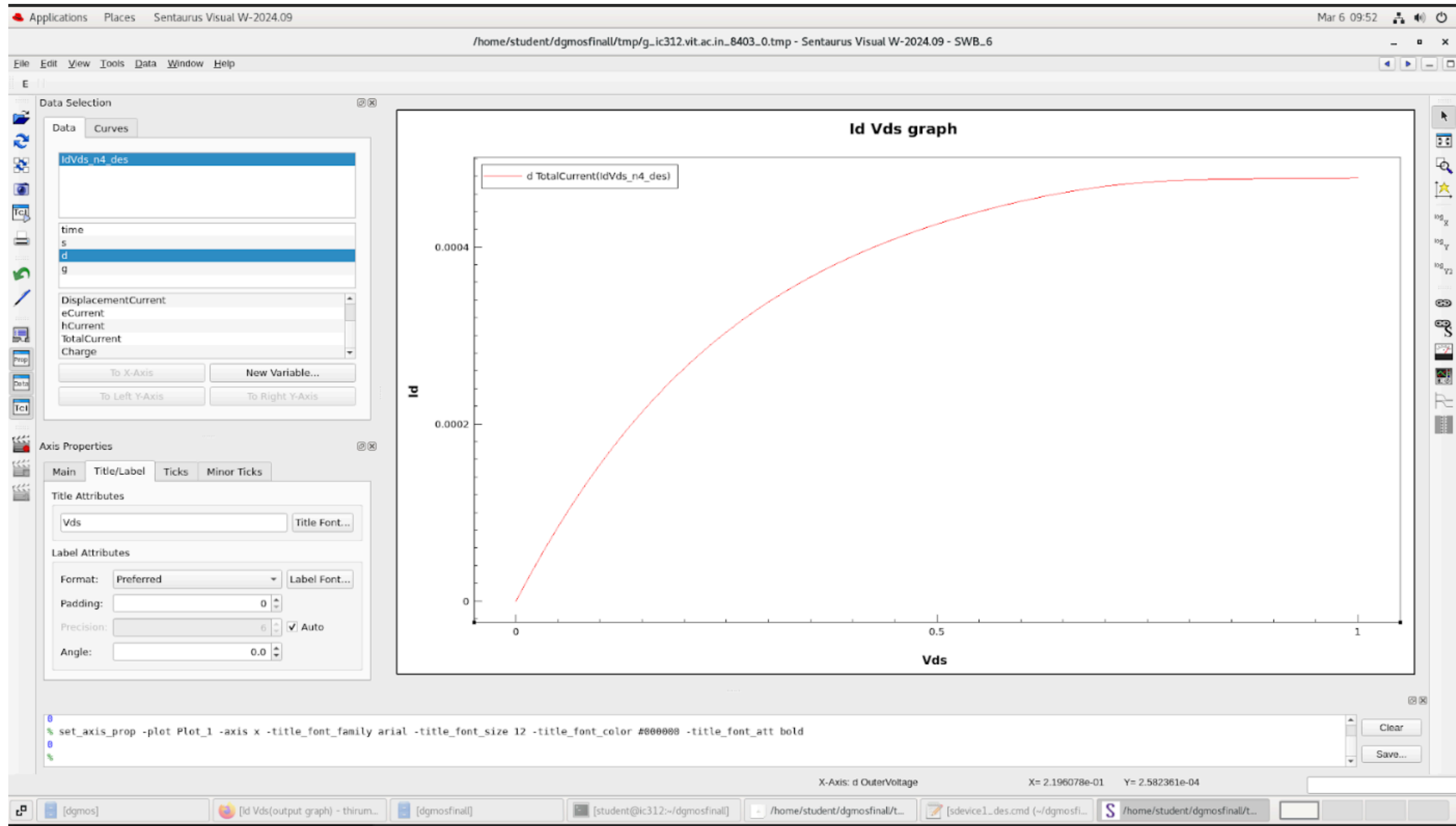


GRAPHS

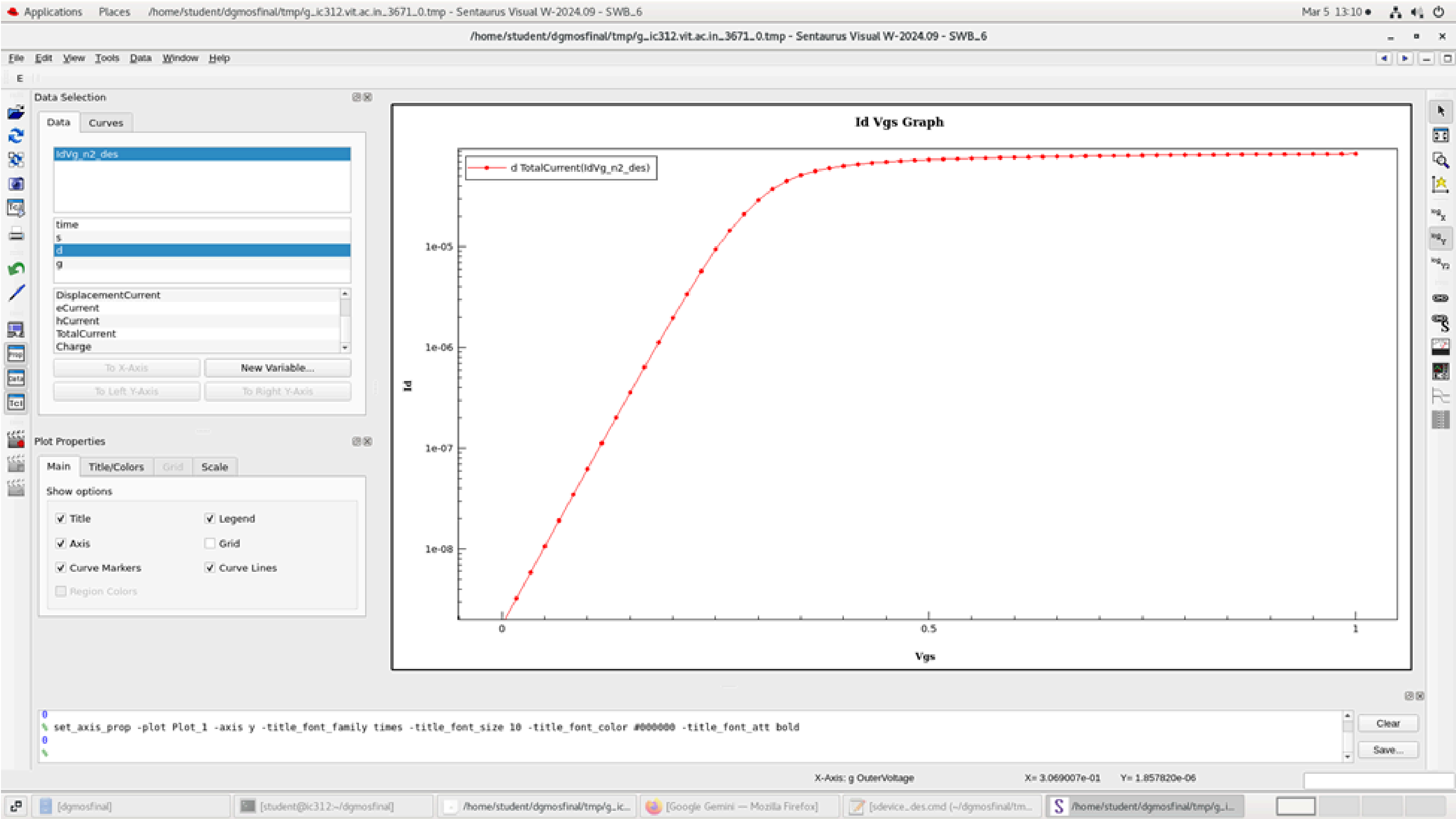
I_d vs V_{gs} (Transfer Characteristics)



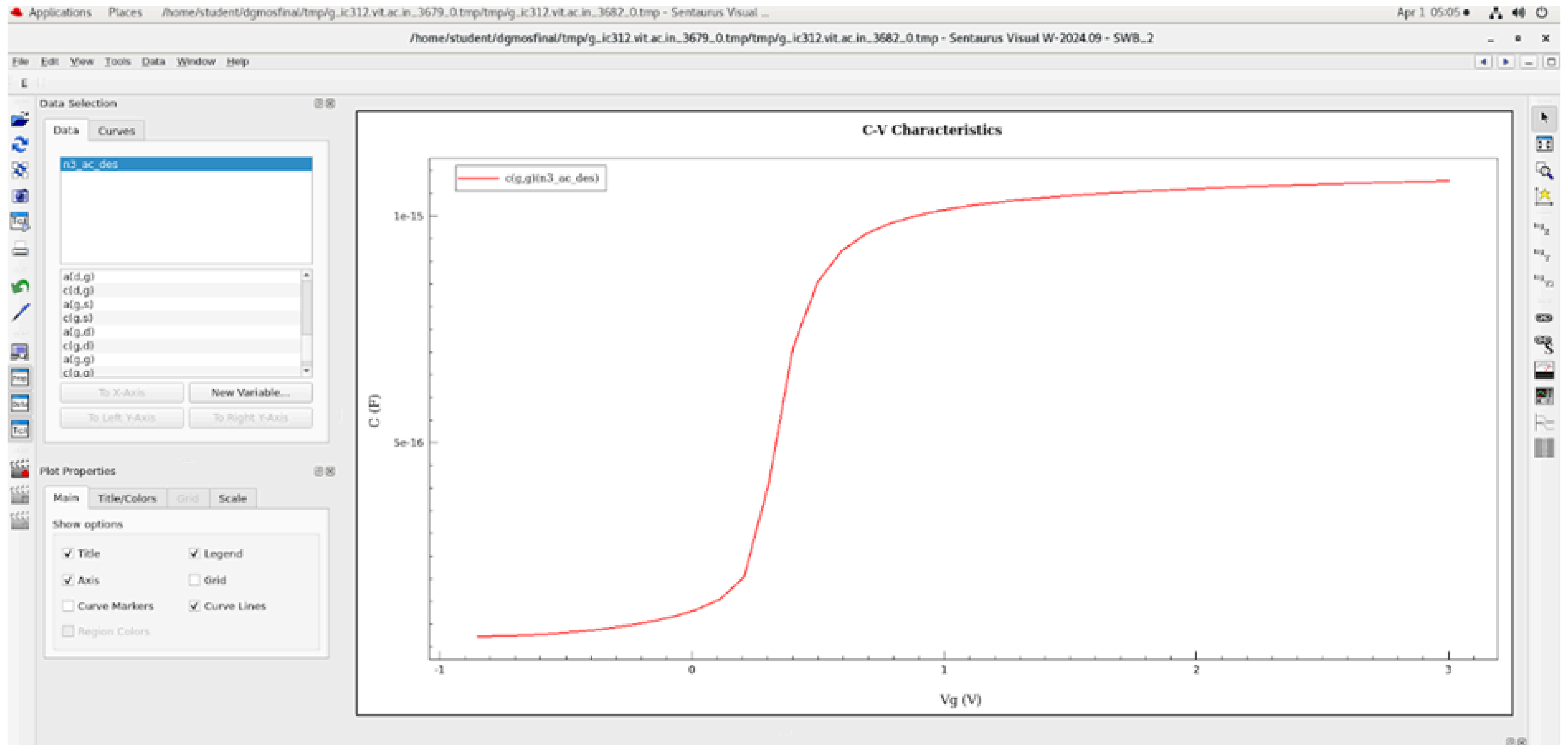
Id vs Vds(Output Characteristics)



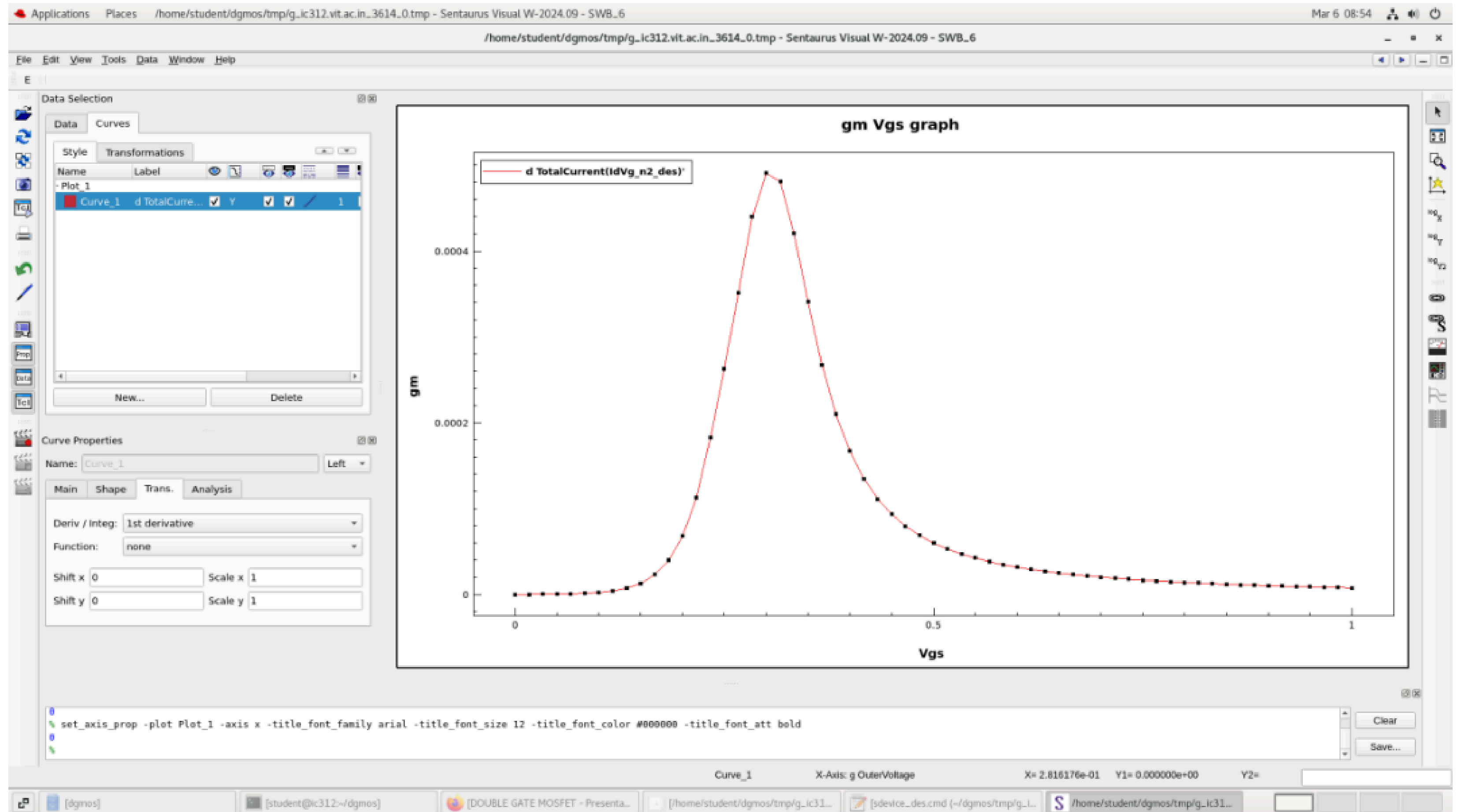
Id vs Vgs(Log graph)



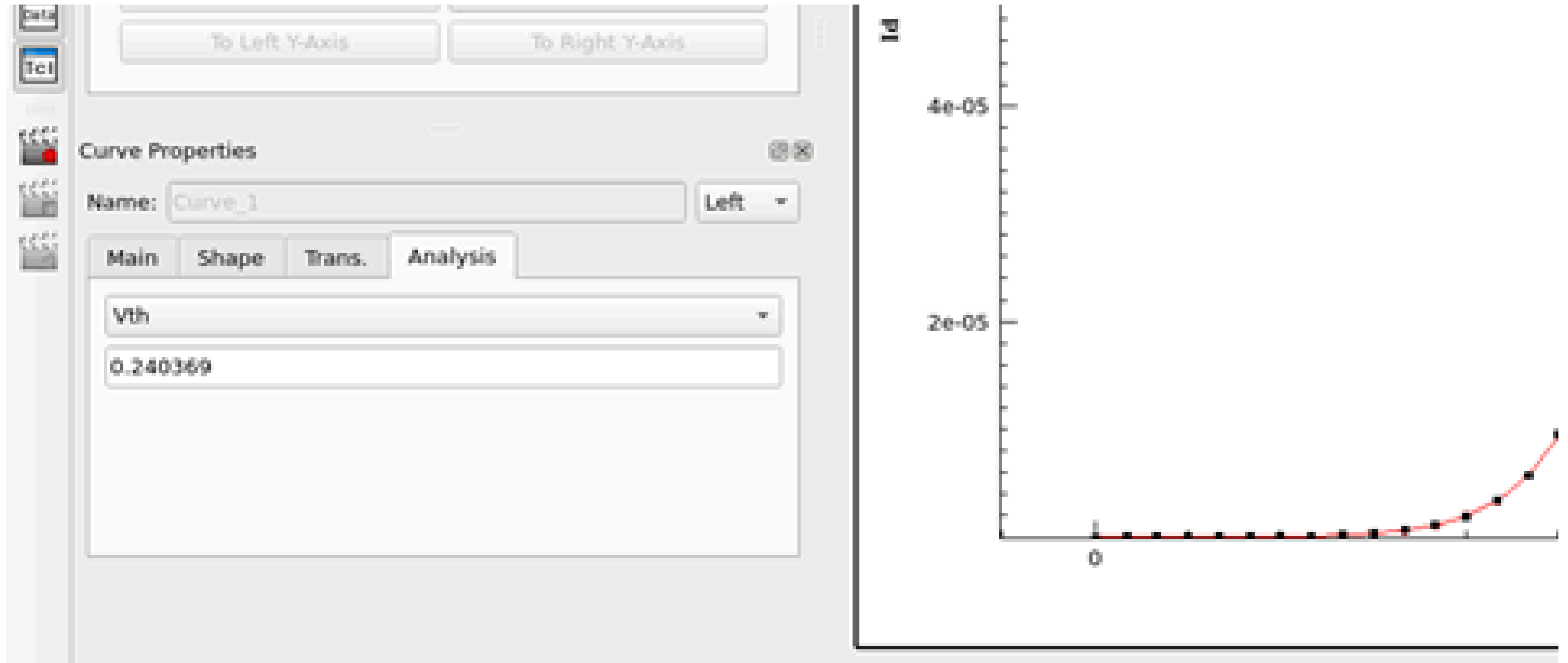
C_{gg} vs V_{gs}(CV Characteristics)



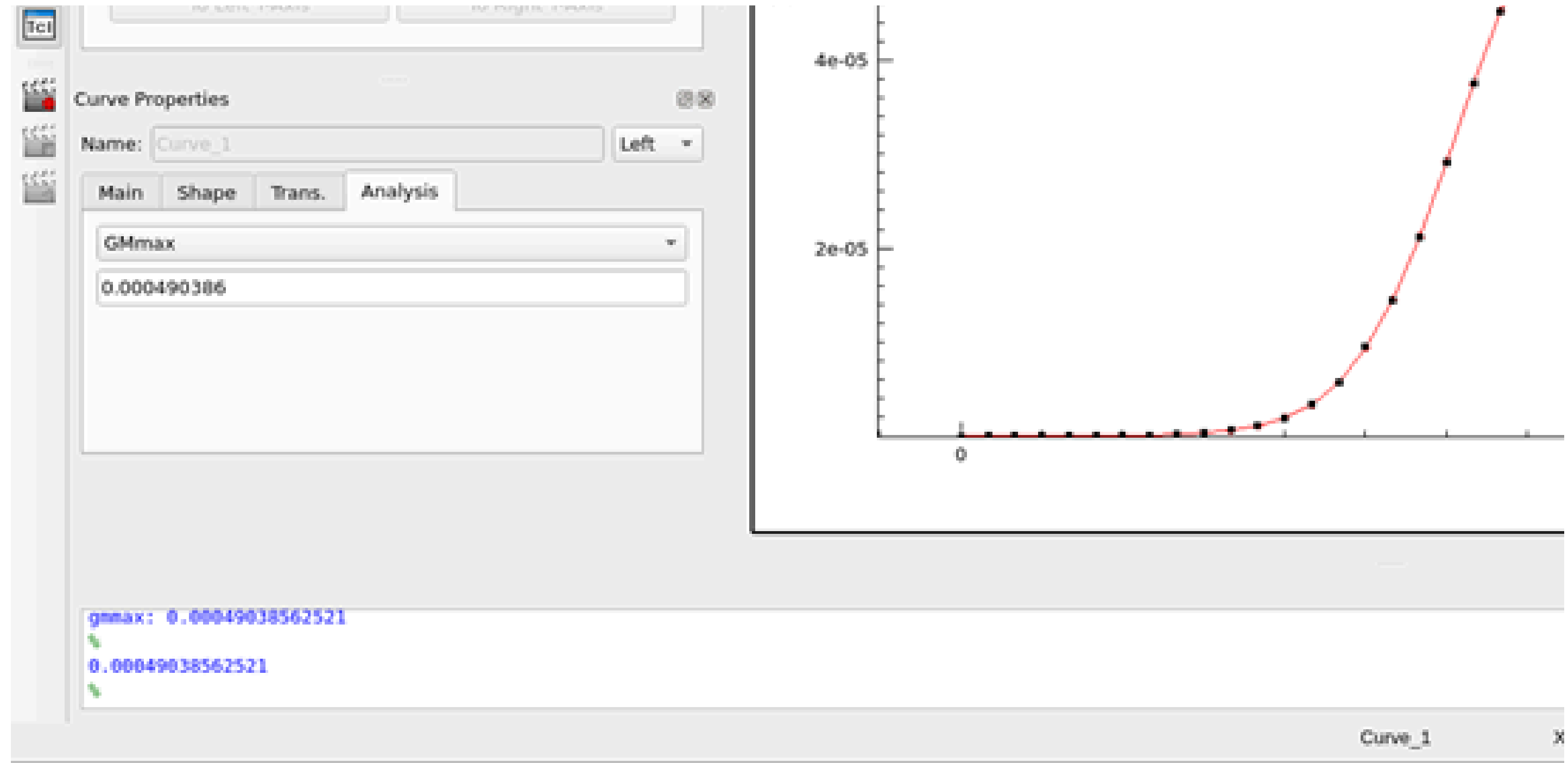
gm vs Vgs(Transconductance Characteristics)



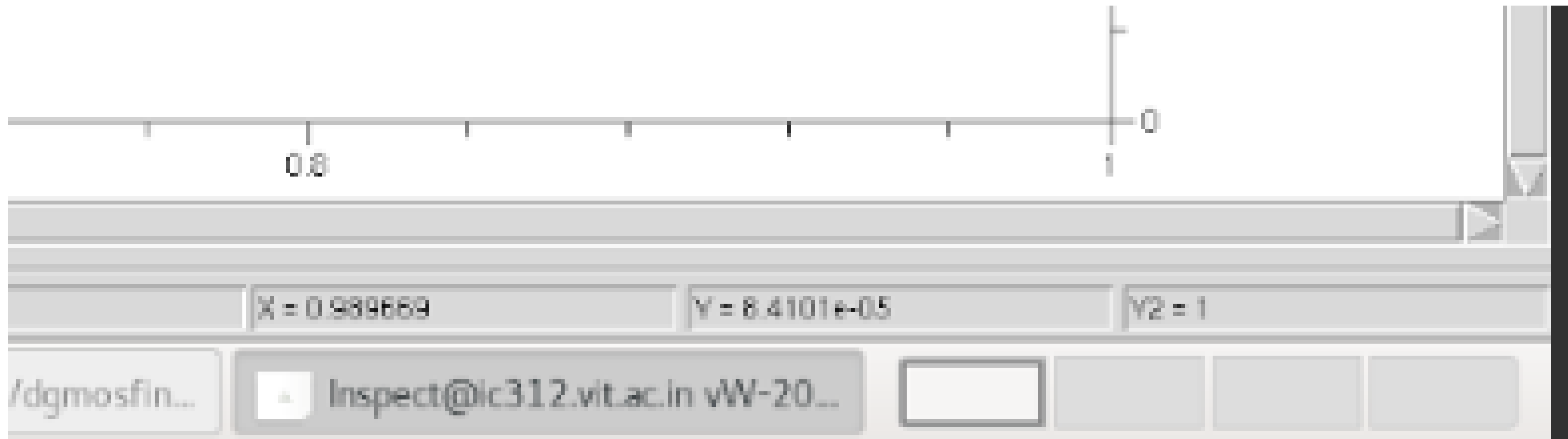
$$V_{th} = 0.240369 \text{ V}$$



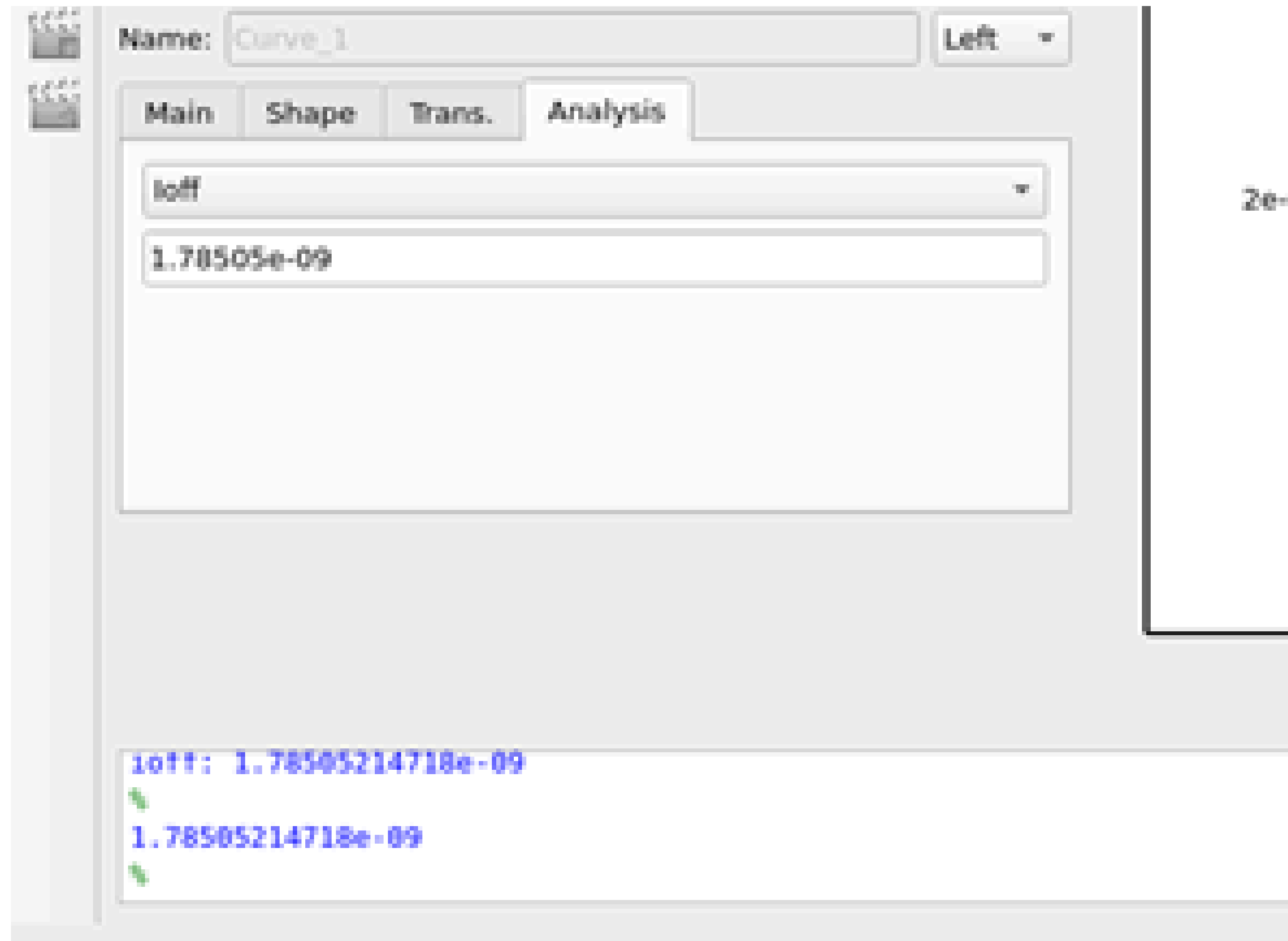
$$g_m = 0.490386 \text{ mA/V}$$



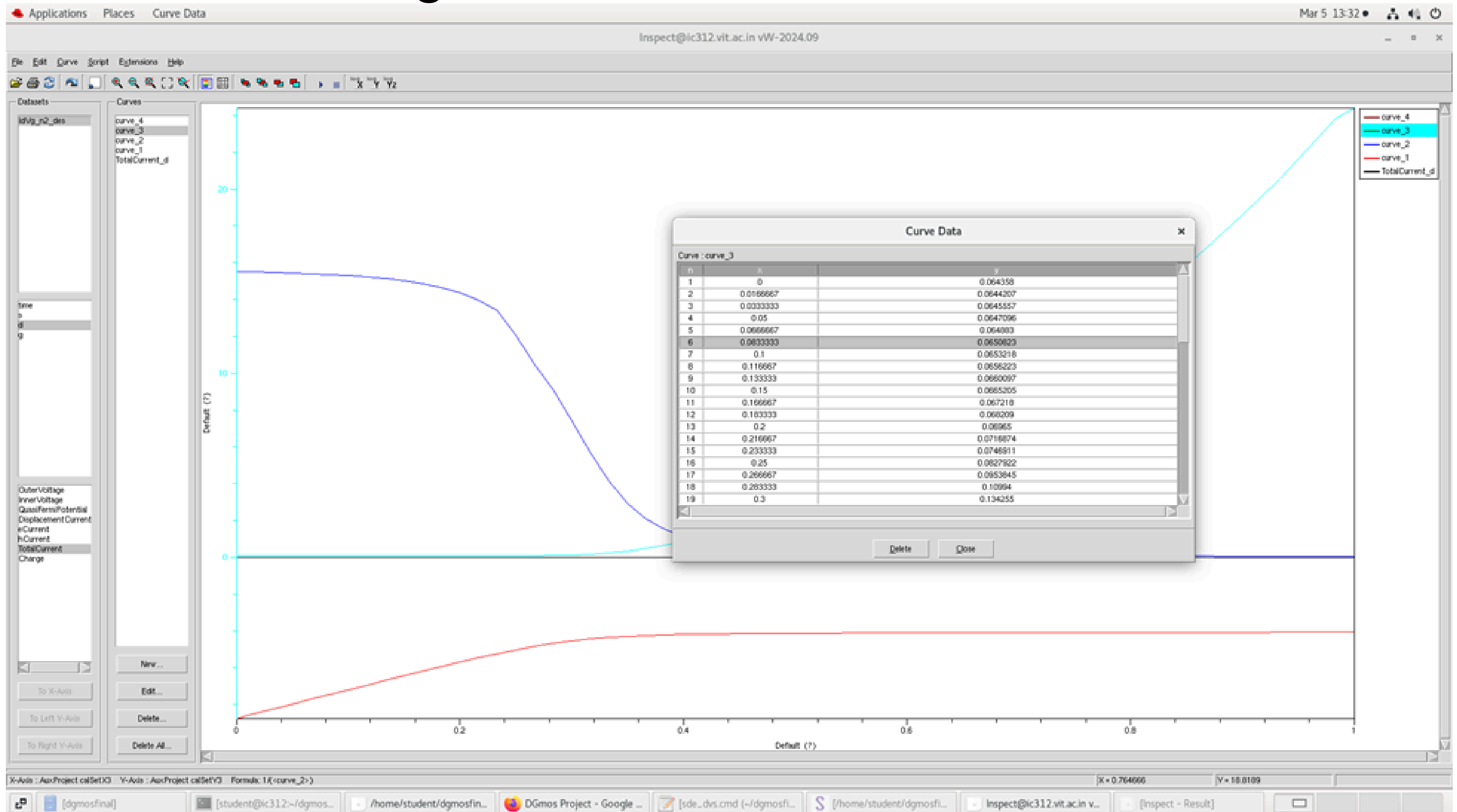
$$I_{on} = 8.4101 \times 10^{-5} \text{ A}$$



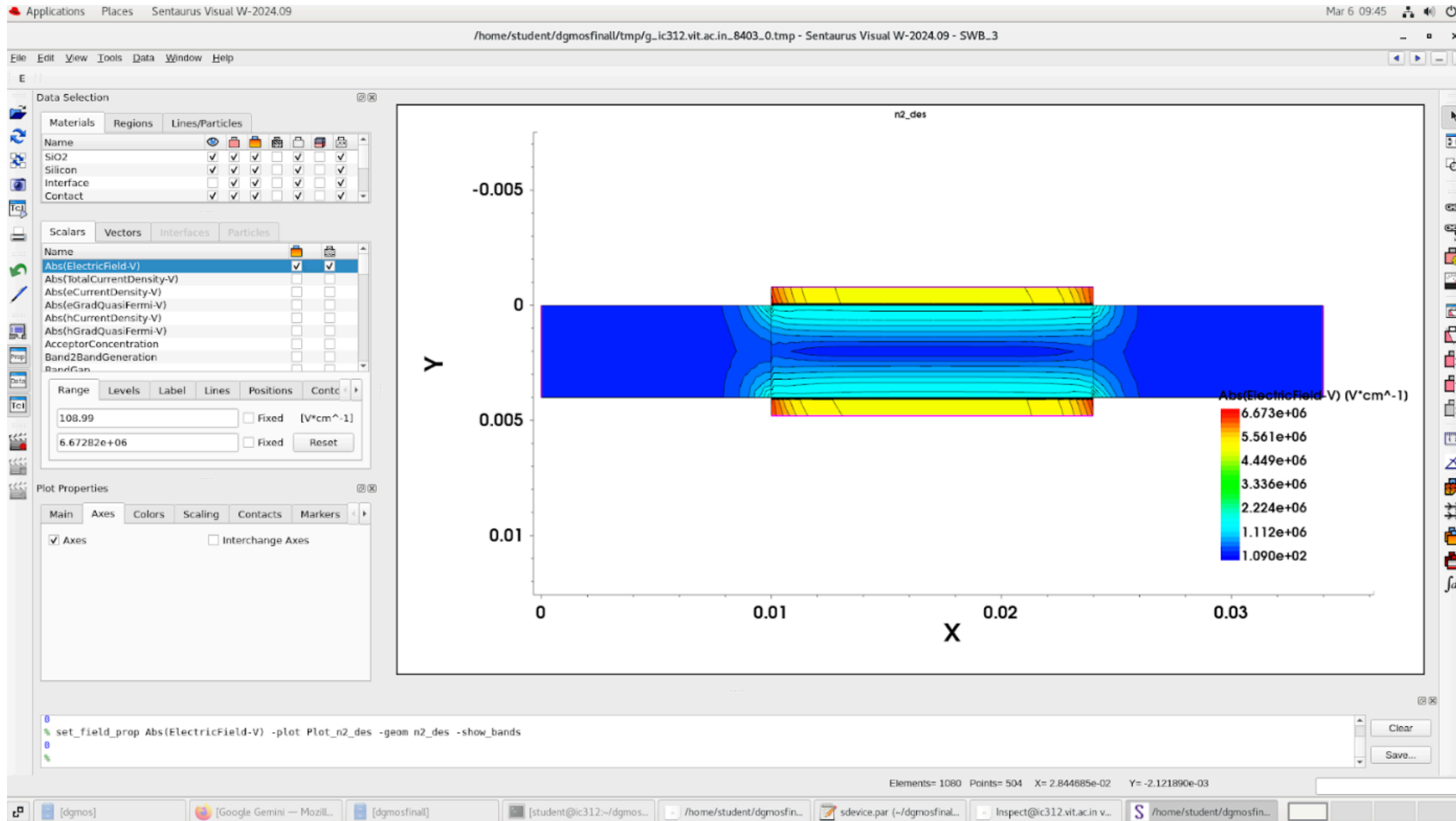
$$I_{\text{off}} = 1.7805 \times 10^{-9} \text{ A}$$



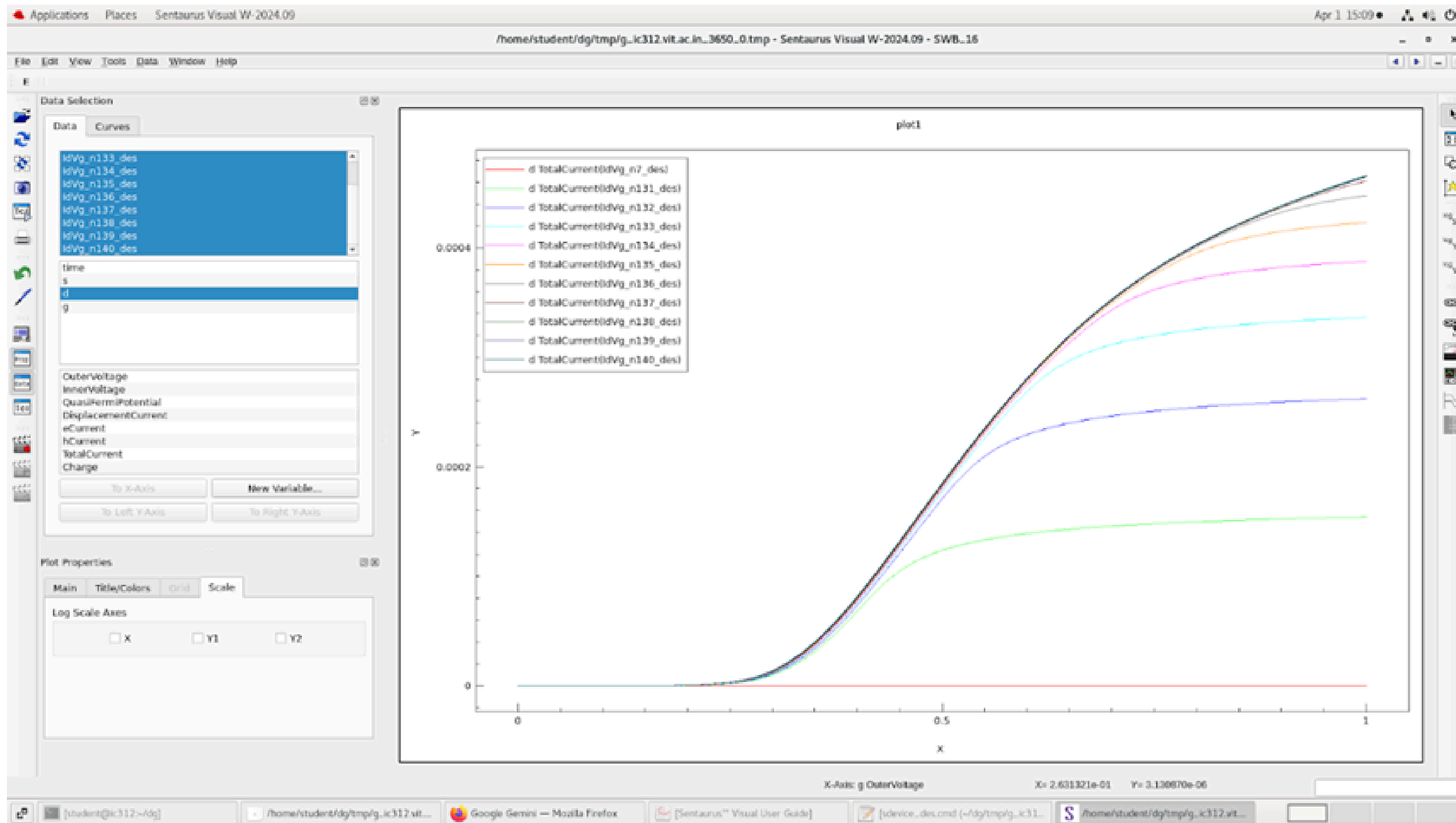
Subthreshold Swing = 65mV/dec



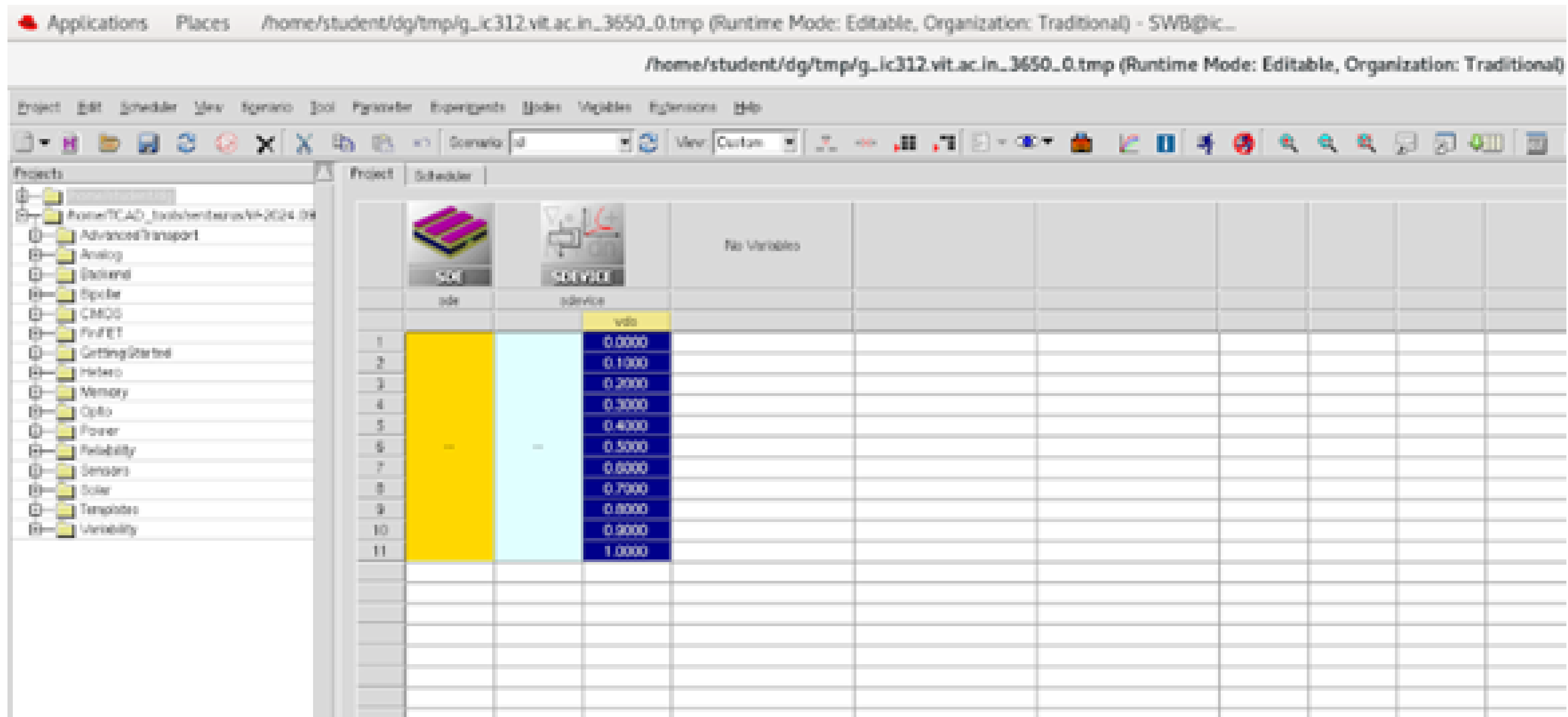
Electric Field:



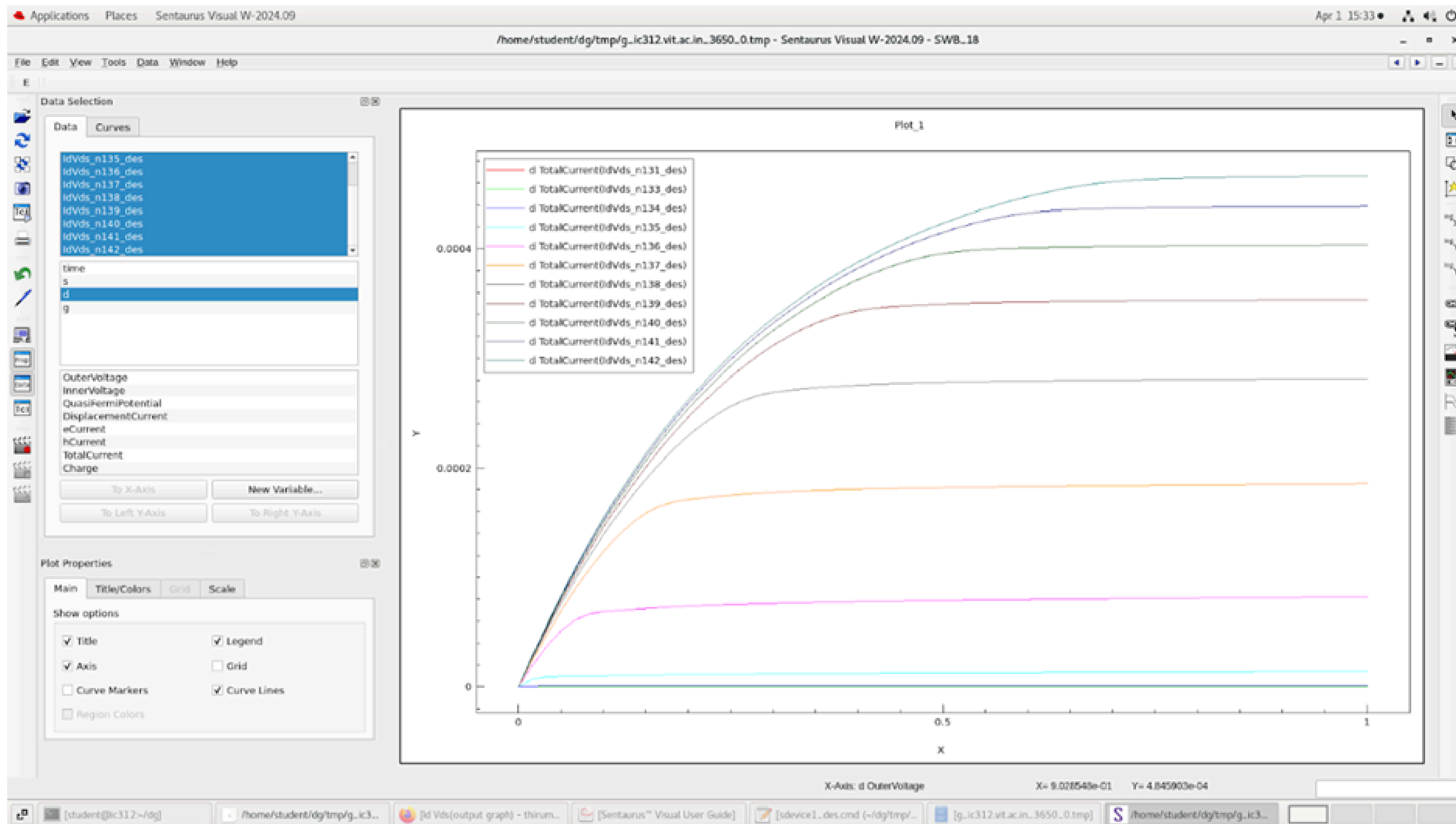
Id-Vgs PARAMETRIC ANALYSIS(VARYING Vds)



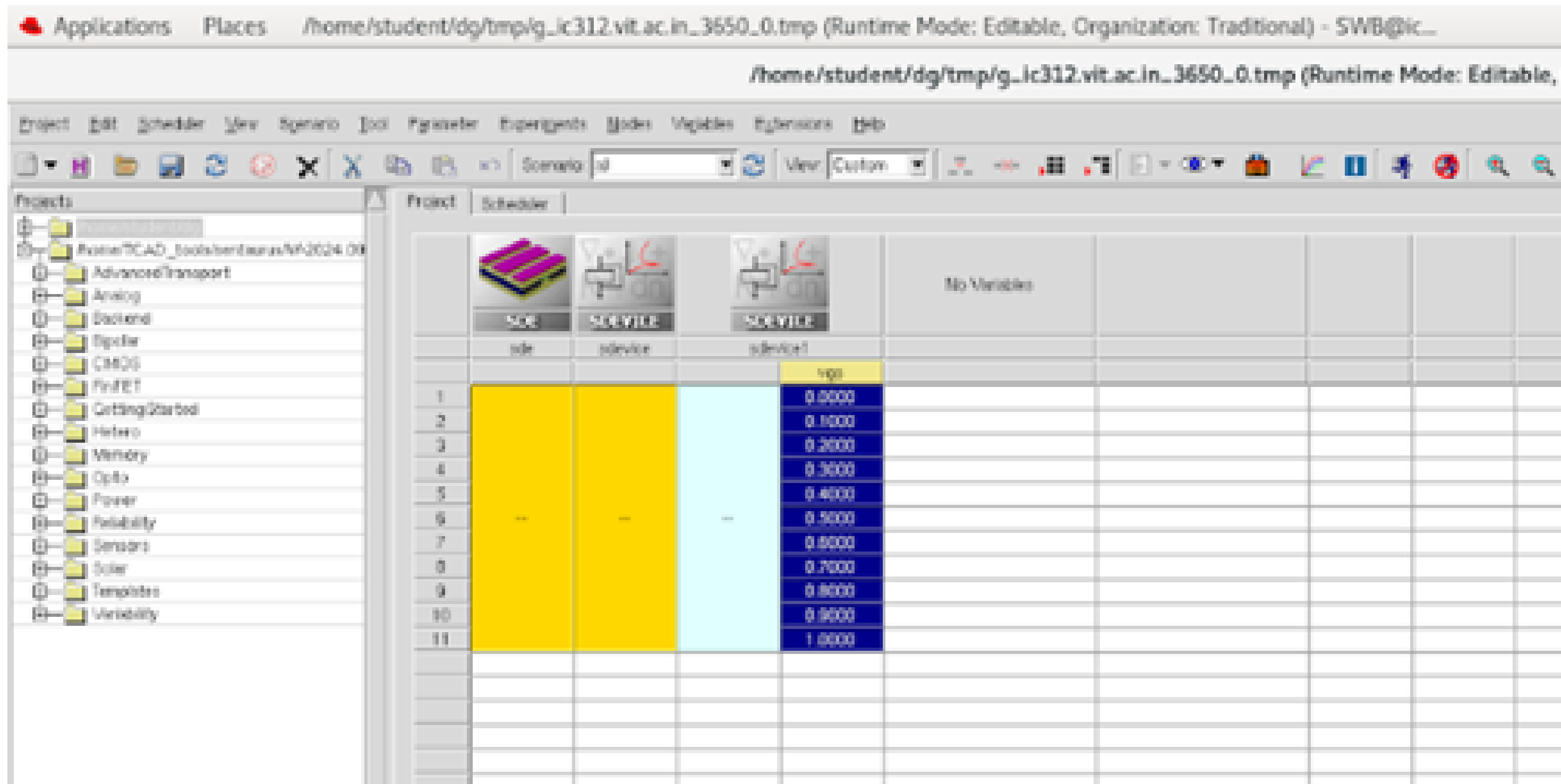
VARYING V_{ds} IN I_d - V_{gs} GRAPH(SWB)



Id-Vds PARAMETRIC ANALYSIS(VARYING V_{gs})



VARYING V_{gs} IN I_d - V_{ds} GRAPH(SWB)



INTEGRATING SENTAURUS WITH CADENCE:

1)Run DC Simulation

- Use Sentaurus Device
- Perform nested sweep:
- Sweep Vgs in Id-Vds graph or Vds in Id-Vgs graph
- Obtain Id data as a function of Vds and Vgs

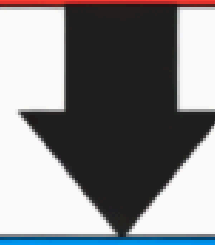
2) Export Data

- Export results as CSV
- Ensure format contains: Vgs, Vds, Id
- Convert CSV to VAMS(Verilog Analog and mixed signal) format

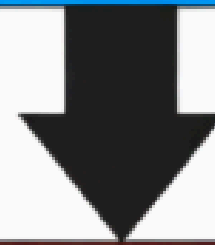
3)Create Verilog-A LUT Model

- In Cadence Virtuoso,use VAMS file LUT and create symbol
- Two VAMS file LUTs are used-NMOSLUT (for dc analysis) and CGGLUT(for ac analysis and transient analysis)-so create one symbol for each and then create amplifier schematic

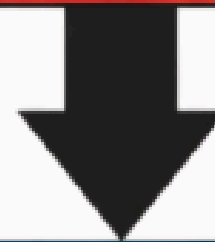
Device simulation in TCAD



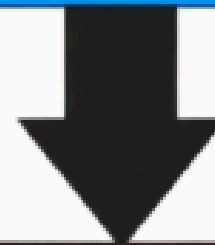
Extraction of device characteristics
from TCAD simulation



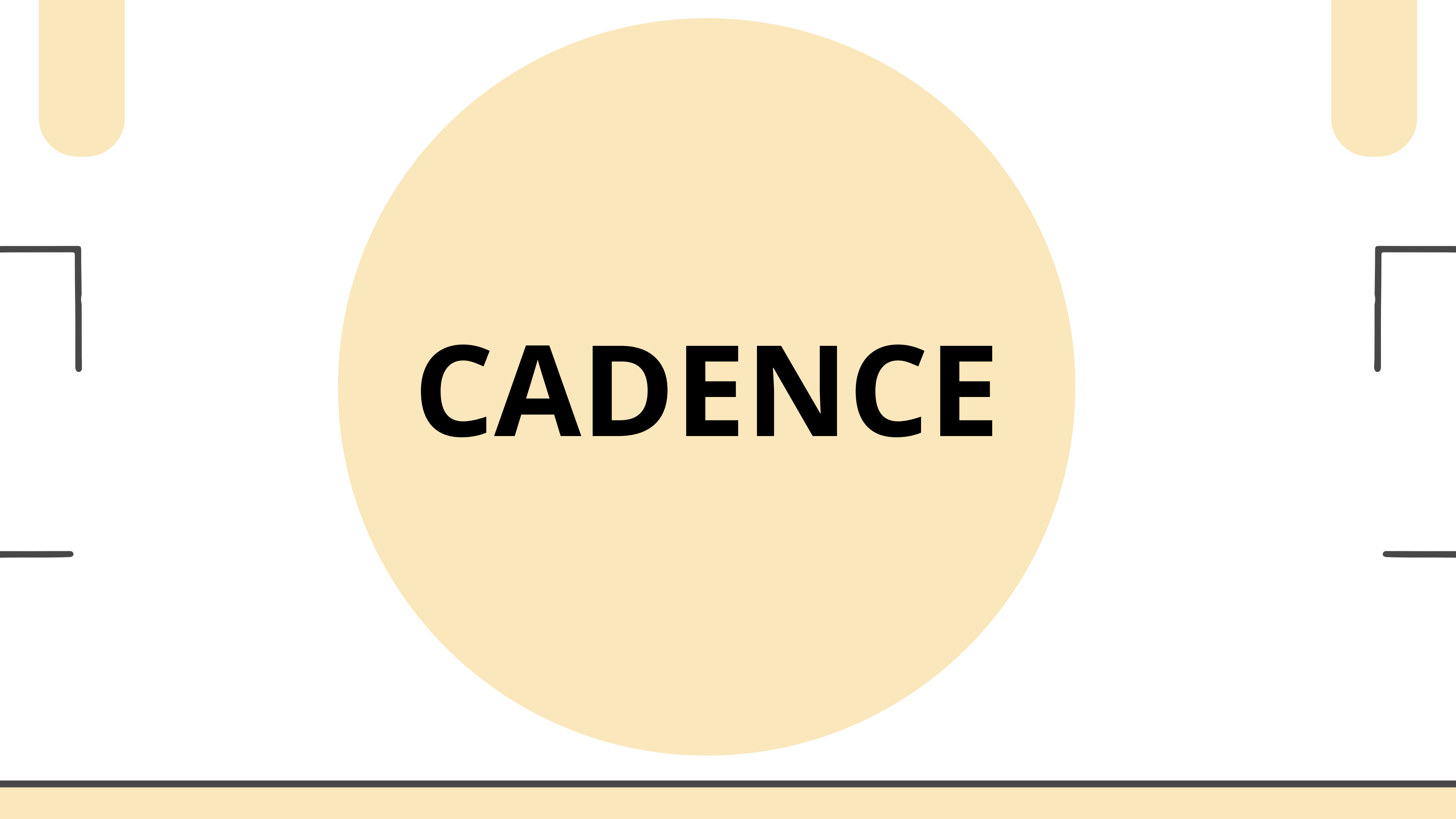
Prepartion of LUT from extracted data



Modelling of devices using LUT's in VerilogA

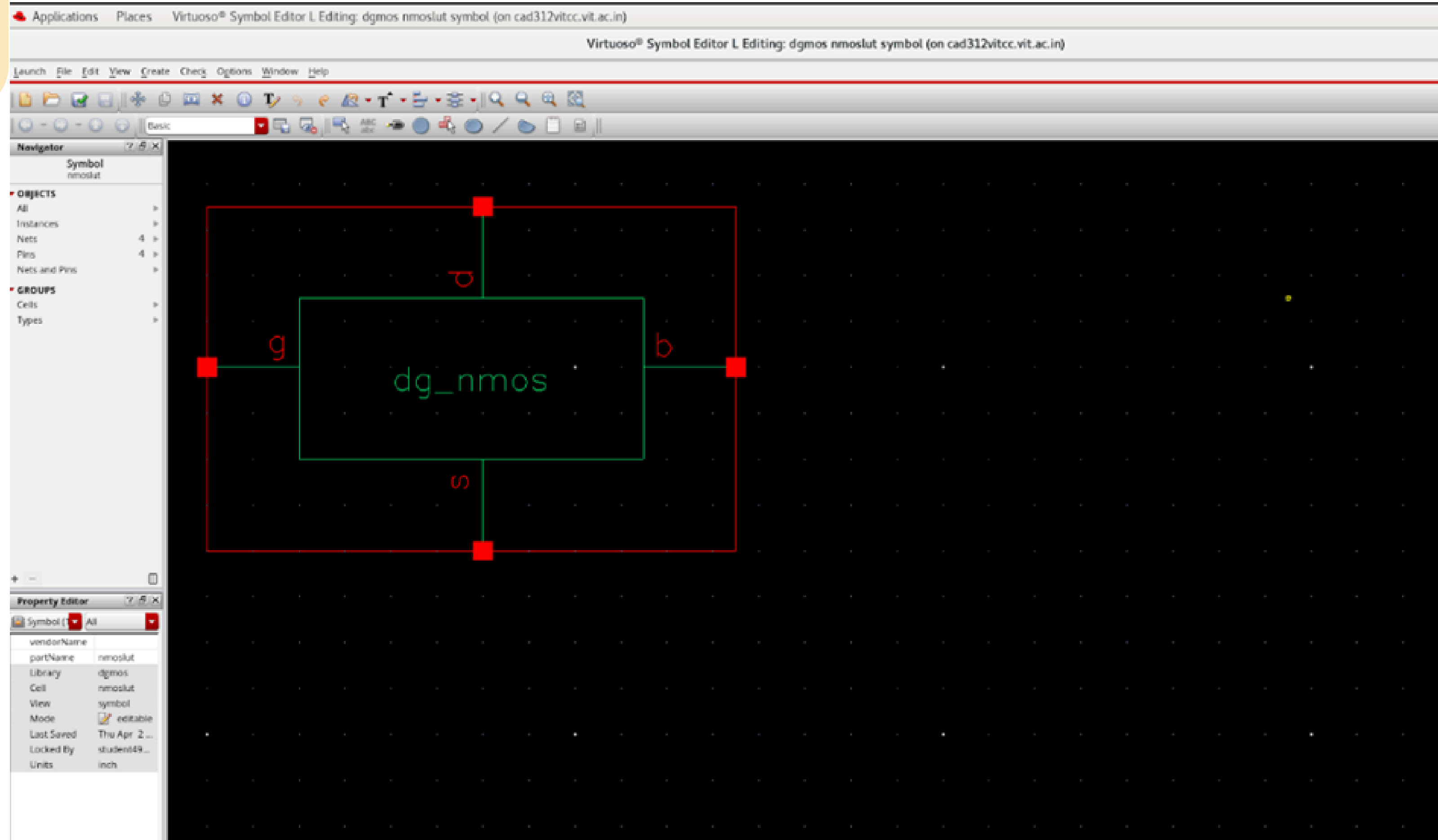


Implementation of amplifier using
Cadence Virtuoso

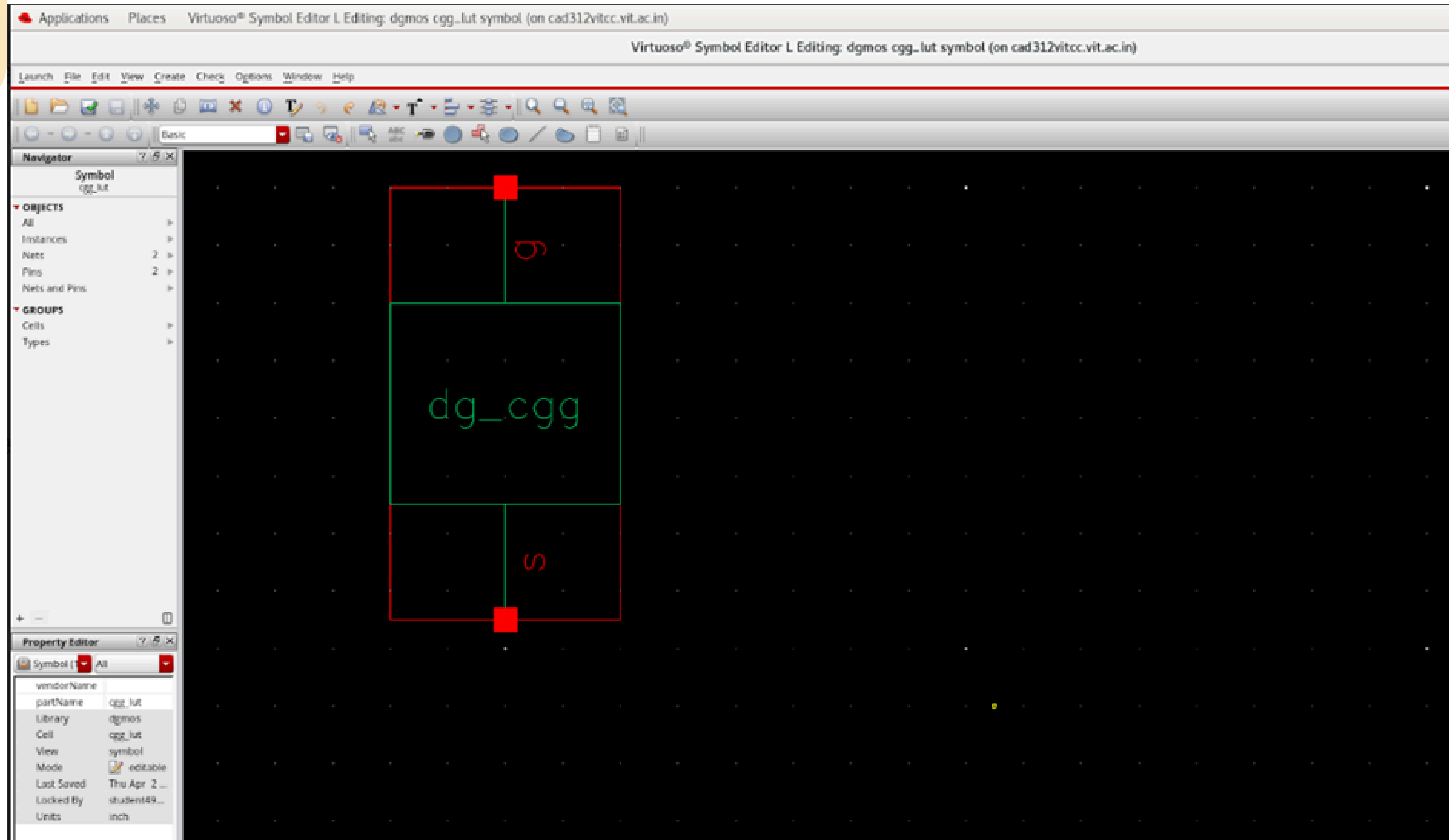


CADENCE

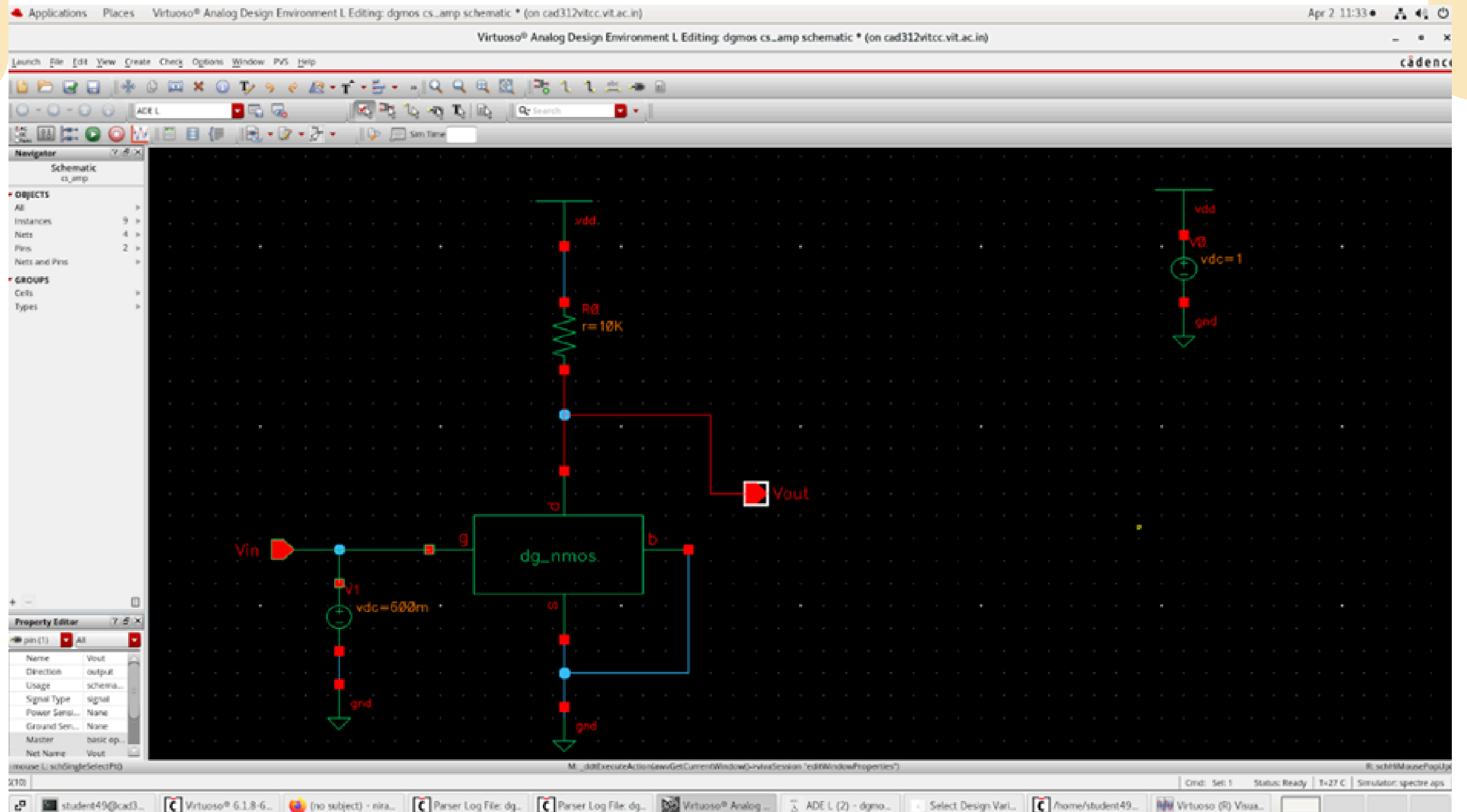
NMOS LUT SYMBOL



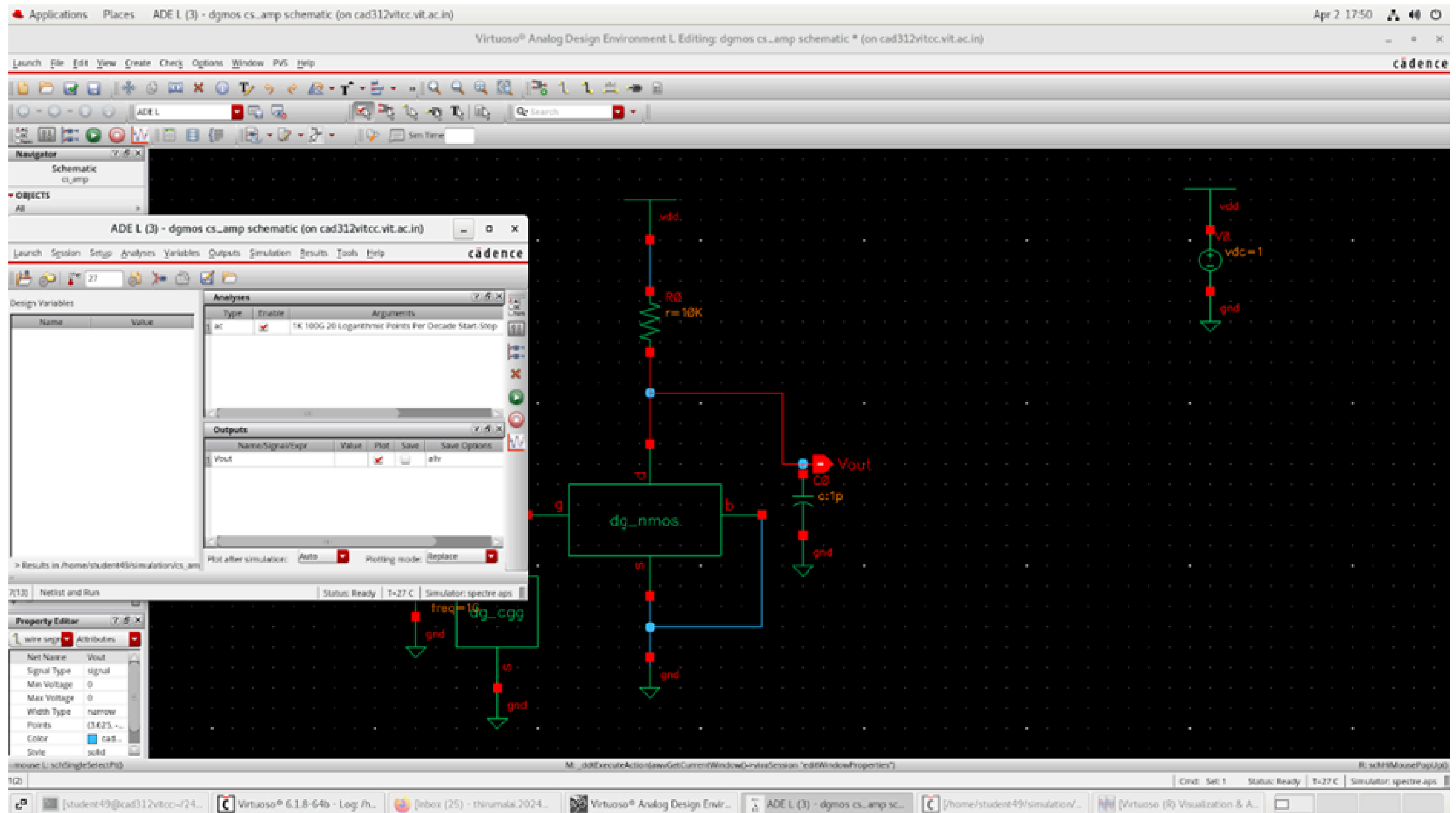
CGG LUT SYMBOL



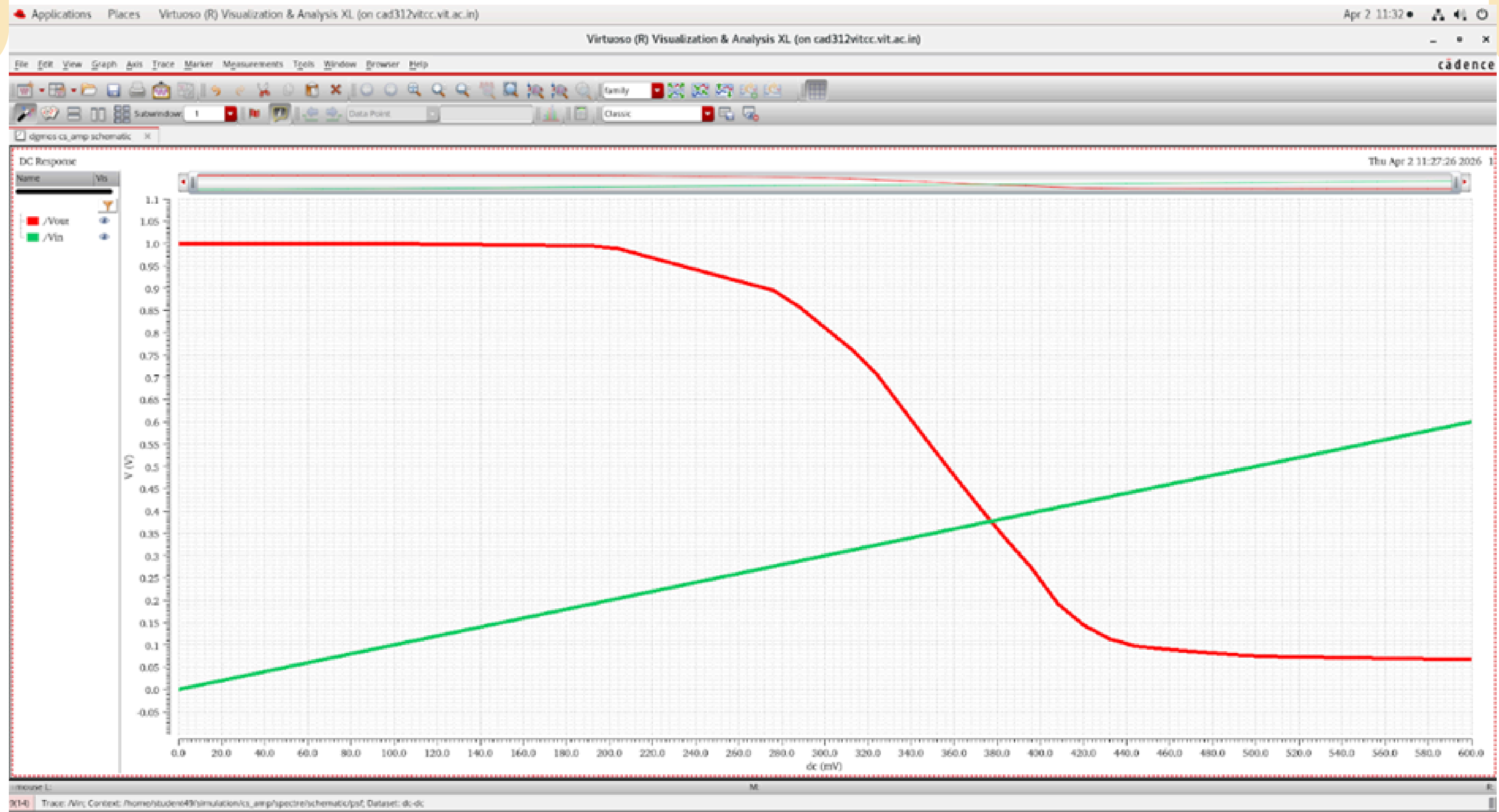
AMPLIFIER SCHEMATIC



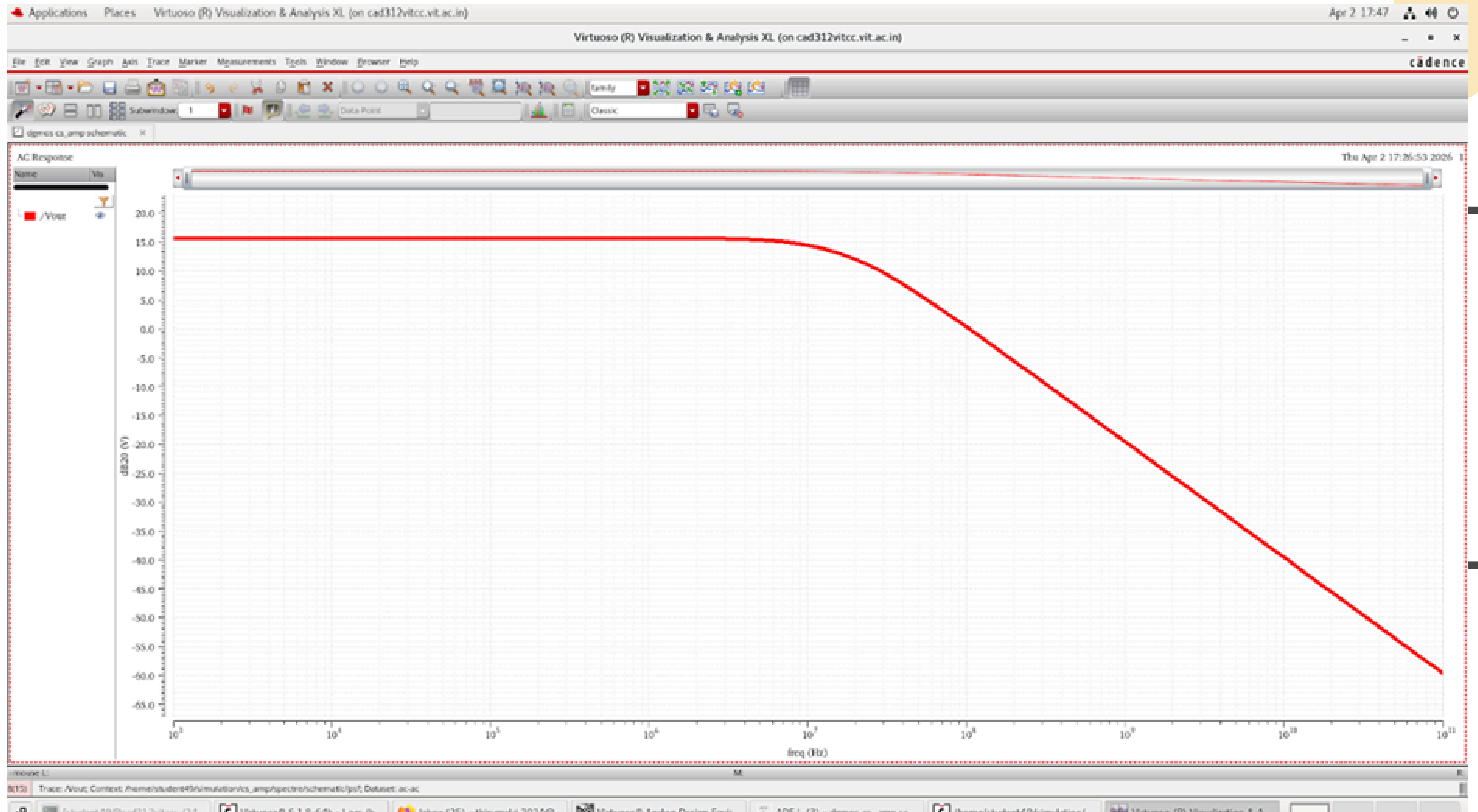
AMPLIFIER SCHEMATIC



DC ANALYSIS(VOLTAGE TRANSFER CHARACTERISTICS)



AC ANALYSIS



TRANSIENT ANALYSIS



PARAMETERS FROM DC AND AC ANALYSIS

DC Analysis

Parameter	Value
DC Operating Point (V_{in} bias)	357 mV
V_{out} at operating point	380 mV
V_{in} sweep range	0 to 600 mV
V_{out} high (cutoff region)	1.0 V
V_{out} low (deep triode)	70 mV
Transition region	220 mV to 430 mV
Crossover ($V_{in} = V_{out}$)	390 mV

AC Analysis

Parameter	Value
Midband Gain	15.5 dB
Midband Gain (linear)	5.96 V/V
3dB Bandwidth	5 MHz
Gain at -3dB	12.5 dB
Gain-Bandwidth Product	30 MHz
Roll-off slope	-20 dB/decade
Flat band range	10 Hz to 1 MHz

PARAMETERS FROM TRANSIENT ANALYSIS

Transient Analysis

Parameter	Value
DC Bias (Vin)	357 mV
DC Bias (Vout)	500 mV
Vin peak	380 mV
Vin peak-to-peak	40 mV
Vout peak	620 mV
Vout peak-to-peak	240 mV
Voltage Gain	6 V/V
Voltage Gain (dB)	15.56 dB
Signal Frequency	1 MHz
Phase shift	180°



THANK YOU